Parallel Deep Learning for NLOS Detection and Error Mitigation in UWB Positioning

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***Abstract*—Ultrawideband (UWB) technology is extensively applied in indoor high-precision localization scenarios. However, barriers along the radio signal path can lead to nonline-of-sight (NLOS) propagation, thereby reducing positioning reliability. Therefore, identifying NLOS conditions and mitigating the associated errors is essential. In this article, a novel NLOS detection and mitigation approach is introduced, leveraging the parallel spatiotemporal feature fusion network (PSTFFN) and spatiotemporal attention regression network (STARN). We utilize continuous wavelet transform to convert the 1-D channel impulse response (CIR) signal into a 2-D time–frequency diagram of CIR image data. By incorporating an attention mechanism and handcrafted features, PSTFFN enhances its ability to differentiate between scenarios with similar features. Based on the NLOS classification results, the proposed STARN model, tailored to specific scenarios, further mitigates NLOS errors. Experimental results indicate that the proposed method performs highly in both binary and multiclass NLOS classification, with accuracy rates exceeding 96.74%. Moreover, using this approach, the root-mean-square error (RMSE) and mean absolute error (MAE) of distance measurements are reduced from 168.04 cm and 106.86 cm to 7.54 cm and 7.78 cm, respectively. In real-world indoor positioning experiments, this method improves positioning accuracy by over 81.86% and achieves real-time response performance.**

***Index Terms*—Error mitigation, indoor positioning system, nonline-of-sight (NLOS) detection, parallel deep learning (DL), ultrawideband (UWB).**

I. Introduction

In THE era of the Internet of Things (IoT), precise positioning technology has become crucial for various applications, from navigation systems to smart home automation [1]. While GPS and BeiDou have revolutionized outdoor positioning, satellite signals cannot penetrate walls and obstacles due to signal attenuation and reflection, limiting their use inside buildings [2]. Traditional indoor positioning systems rely on wireless communication technologies, such as infrared, Bluetooth, Wi-Fi, and radio frequency identification (RFID) [3]. Despite significant advancements in indoor positioning with these technologies, inherent limitations in range and accuracy prevent them from meeting the growing demand for precise indoor positioning. This has led researchers and industry professionals to explore more robust alternatives.

Ultrawideband (UWB) technology is a promising solution for overcoming the limitations of traditional RF technologies in indoor positioning. Operating across a broad frequency range, UWB enables high-speed, low-power data transmission, with its wide bandwidth providing fine time resolution to enhance distance measurement accuracy [4]. These properties make it ideal for indoor environments, driving its widespread adoption in consumer electronics. Major brands like Apple and Samsung are now integrating UWB into smartphones, advancing location-based services and applications [5].

Despite the advantages of UWB, its indoor localization systems encounter essential challenges, particularly in nonline-of-sight (NLOS) environments. As illustrated in Fig. 1, when signals encounter obstructions like walls, furniture, or people, causing diffraction, reflection, and refraction. Consequently, only a fraction of the signal reaches the receiver directly, resulting in variations in arrival time and phase, known as multipath effects [6]. This is reflected in the channel state information (CSI) as an increase in local peaks. In UWB, precise distance measurements depend on the ideal line-of-sight (LOS) condition for time-of-flight (ToF) calculations. However, in indoor environments where non-LOS (NLOS) is common, ToF estimates experience delays, leading to a positive bias in distance values. Many UWB localization algorithms, such as trilateration [7] and least squares (LSs) [8], assume perfect LOS scenarios, making them susceptible to errors under NLOS and multipath effects. Effective NLOS detection and error mitigation are thus crucial to enhance the robustness of UWB systems and expand their applications in fields, such as healthcare, logistics, and smart buildings.

In recent years, research on mitigating UWB positioning errors has primarily focused on two approaches. The first involves fusing sensor data from sources like LiDAR [9] and inertial measurement unit (IMU) [10] using methods like Kalman filters, federated filtering, and generalized probabilistic data association. This fusion helps to reduce UWB ranging errors caused by NLOS conditions, thereby decreasing overall positioning error. The second approach focuses

Received 21 March 2025; revised 13 June 2025; accepted 6 August 2025. Date of publication 11 August 2025; date of current version 9 October 2025. This work was supported in part by the Natural Science Foundation of Hunan province of China under Grant 2025JJ50270. (*Corresponding author: Ming-Song Chen.*)

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Digital Object Identifier 10.1109/JIOT.2025.3597300

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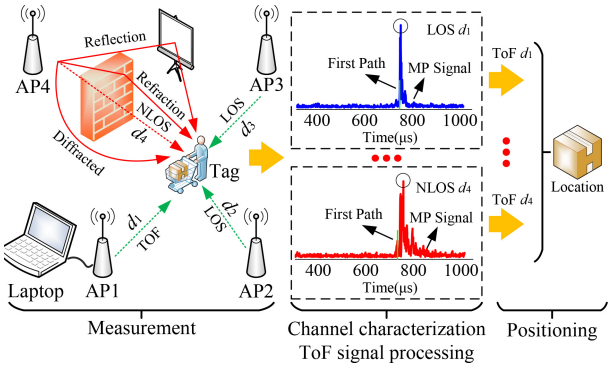


Fig. 1. Diagram of multianchor positioning and NLOS channel characterization.

on accurately detecting NLOS propagation conditions. By selecting nodes with minimal NLOS impact for positioning calculations or mitigating distance measurement errors based on NLOS classification results, this method indirectly reduces positioning errors. Between the two methods, the former may not always lead to significant improvements in positioning accuracy due to sensor precision differences and fusion complexity. The latter has gained attention in recent years for its accuracy, and broader application potential. Common NLOS detection methods, such as hypothesis testing [11], machine learning (ML) [5], and deep learning (DL) [12] techniques. SHT assesses the probability of NLOS conditions by analyzing the statistical characteristics of received signal indicators, evaluating signal feature distributions [11]. ML algorithms extract handcrafted features from UWB channel impulse response (CIR) signals, employing techniques like support vector machines (SVMs) and ensemble classifiers to train models on labeled datasets for distinguishing LOS and NLOS scenarios [5]. DL techniques, such as recurrent neural networks (RNNs) [13] and convolutional neural networks (CNNs) [14], process original CIR signals directly to learn layered features that capture intricate spatiotemporal dependencies. CNNs, for instance, extract spatial features via convolutional layers, whereas RNNs, especially long–short-term memory (LSTM) networks [15], extract sequential dependencies, which makes them ideal for the inherent time-series nature of CIR signals. However, current NLOS detection research largely focuses on binary classification tasks and lacks strong generalization capabilities. Additionally, most studies focus on NLOS detection without integrating NLOS error mitigation strategies.

In light of these considerations, this article proposes a novel NLOS detection method, the parallel spatiotemporal feature fusion network (PSTFFN), which integrates CIR signals, their time–frequency representations, and five handcrafted features to identify specific NLOS occlusion types. Additionally, a targeted NLOS error mitigation approach is introduced, named the spatiotemporal attention regression network (STARN). Based on the classification results from NLOS detection, STARN utilizes CIR signals and seven handcrafted features to predict and mitigate UWB ranging errors. The approach aims to reduce positive UWB ToF ranging errors, ultimately addressing UWB positioning inaccuracies.

The main contributions are as follows.

1) A PSTFFN that fuses UWB CIR data, time–frequency images, and handcrafted features through parallel inputs is proposed for NLOS detection, allowing the model to capture richer feature representations and improving its robustness in detecting signal occlusions.
2) An NLOS error regression prediction model based on the NLOS detection results is proposed, using CIR data and handcrafted features to accurately predict NLOS-induced ranging errors. This approach mitigates localization errors at the source, significantly enhancing positioning accuracy under NLOS conditions.
3) The mean-squared error (MSE) loss function is applied to optimize the regression model, effectively reducing large prediction errors and further improving the overall performance of the model in NLOS environments.

The remainder of this article is structured as follows. Section II offers a brief review of prior work on NLOS detection and error mitigation in UWB systems. Section III introduces the background information on UWB NLOS detection, including preliminary knowledge on time–frequency diagram acquisition, and details the PSTFFN model for NLOS detection and the STARN model for ranging error prediction and mitigation. Section IV discusses the experimental setup, results, and analysis. Finally, Section V wraps up the article and outlines potential future directions.

II. Related Work

A. UWB NLOS Identification

In recent years, NLOS identification research has progressed from traditional ML-based binary classification to advanced DL and multiclass methods, driven by the need for greater robustness and adaptability in complex environments.

For NLOS binary classification, ML methods have long been the mainstream due to their simplicity and interpretability. These approaches typically extract handcrafted features from the UWB channel for classification. For example, Liu et al. [16] proposed an SVM-based classifier that selects features with low computational complexity and high recognition accuracy, improving both speed and performance. Fan and Awan [17] utilized an expectation maximization Gaussian mixture model (EM-GMM) to assign probabilities to LOS and NLOS conditions for each signal. However, ML relies on manual feature engineering, limiting their ability to capture representative features in high-dimensional or complex environments.

To address these limitations, DL techniques have been increasingly applied to NLOS identification tasks. DL models excel at learning complex patterns and automatically extracting high-level features from raw UWB channel data. Cui et al. [18] proposed an NLOS/LOS recognition technique using complex Morlet wavelet transform and CNN (MWT-CNN). MWT-CNN converts the UWB signal into a time–frequency spectrum, leveraging detailed time–frequency features for NLOS identification. Pei et al. [19] developed the fully convolutional network (FCN)-ATTENTION algorithm, which incorporates self-attention mechanisms within a FCN to enhance feature representation and classification accuracy. Li et al. [20] further improved robustness by combining CNN, bidirectional LSTM

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(BiLSTM), and transfer learning, allowing the model to adapt to new environments with limited labeled data.

Recently, research interest has shifted from simple binary NLOS identification to more fine-grained multiclass NLOS classification. Multiclass approaches enable the system to capture a more detailed understanding of the propagation conditions and facilitate more effective ranging error mitigation [21]. For instance, Li et al. [22] proposed a multiclass NLOS identification method based on ensemble learning with an XGBoost classifier. Kim et al. [15] employed an LSTM-based architecture for time-series CIR classification across multiple channel conditions. Tu et al. [23] utilized a hybrid CNN+LSTM model to distinguish between different types of NLOS. Additionally, Wang et al. [14] constructed the CNN-A model, which applies 1-D wavelet packet analysis combined with CNNs to extract frequency-domain features, thereby enhancing the model's multiclassification capability for various NLOS channel types.

To further exploit channel information and improve generalization, multiinput and parallel DL architectures have gained traction. These architectures enable simultaneous processing and fusion of different input features, leading to more robust NLOS multiclass identification. Deng et al. [24] proposed a novel NLOS recognition and mitigation approach using a multiinput parallel DL model and Gramian angle field. Wei et al. [25] developed a multiple input learning (MIL) neural network handling both CIR and time–frequency diagram of CIR (TFDOCIR), better capturing the diversity of signal characteristics and achieving improved classification over conventional single-input networks.

B. NLOS Error Mitigation

After identifying the NLOS propagation state, it is necessary to further mitigate localization errors. Traditional methods address this by using Kalman filters or factor graph methods to fuse IMU data and directly reduce localization errors. For example, Cui et al. [26] employed the extended Kalman filter (EKF) for target localization based on range estimates.

Although integrating an IMU can enhance the performance of UWB indoor positioning systems in certain scenarios, the overall accuracy is still limited by UWB's precision due to the IMU's inherent drift and accumulated errors. To address this, researchers have proposed two main approaches leveraging NLOS detection. One involves constructing a wireless sensor network that assigns different weights to distance measurements based on NLOS detection results, followed by location estimation through LS calculations [8]. However, when all base stations and tags are in an NLOS state, this method's effectiveness in mitigating positioning errors is greatly reduced.

To address this, some researchers have proposed low-cost methods for directly mitigating ranging errors. For example, Angarano et al. [27] introduced a range error mitigation network, where the network autonomously learns how to compensate for LOS and NLOS ranging errors. The model implicitly performs the NLOS classification during the mitigation process, eliminating the need for an additional classification step. Zhang and Wang [28] utilized a neural process regressor within a network to accurately learn environmental and ranging-related information, directly mitigating NLOS ranging errors. However, in complex indoor scenarios, the effectiveness of directly mitigating ranging errors through network learning is limited. Research in [29] demonstrated the advantages of a two-step ranging error mitigation approach based on NLOS recognition. This method first performs NLOS subclassification using DL and then selects a corresponding error regression model based on the subclassification results. Compared to direct error mitigation methods, this approach significantly improves error reduction.

In conclusion, although DL-based NLOS recognition has gained growing attention, especially in multiclass NLOS identification, existing multiclass models often face challenges in fully utilizing channel characteristics in complex environments due to limited feature extraction and generalization capabilities. Parallel architectures with multifeature inputs show promise in capturing diverse signal properties and enhancing robustness, allowing for more effective correction of specific ranging and positioning errors. Additionally, the mainstream approach for mitigating positioning errors still relies on IMU fusion, which increases costs and provides limited error reduction. Most DL-based studies separate NLOS identification and mitigation, missing the opportunity to leverage identification results to improve NLOS error mitigation. Integrated approaches are rare and underexplored.

III. System Model and Operational Principle

This section first provides an overview of the UWB physical layer, focusing on the CIR-based mathematical model. Next, it provides a detailed explanation of the principles behind generating time–frequency spectrograms of CIR signals using the continuous wavelet transform (CWT), along with an analysis of the LOS/NLOS CIR signal curves and their corresponding time–frequency spectrograms obtained from experiments. Furthermore, the design and workflow of the PSTFFN are presented. Finally, the section describes the STARN's structure and working principles for mitigating NLOS ranging errors after NLOS occlusion level detection using the PSTFFN.

A. NLOS/LOS CIR Signal in UWB System

The UWB devices DWM1000 used in this study employ direct sequence (DS) spread spectrum technology at the physical layer [30], where a single tag communicates with multiple anchors for distance estimation. DS expands the signal spectrum by performing a modulo-2 addition between the signal $d_{ds}(t)$ and a high-frequency pseudorandom code $PN(t)$, producing the modulated signal $s_{ds}(t)$. At the receiver end, the signal $r_{ds}(t)$ is despread using the same pseudorandom code $PN(t)$ to recover the original signal $d_{ds}(t)$. A simplified diagram of this principle is shown in Fig. 2. Additionally, the DWM1000 utilizes a preamble and start frame delimiter (SFD) to achieve precise pseudorandom code synchronization, counteracting clock drift and mitigating the effects of multipath interference.

Compared to Time-Hopping (TH) [31], DS/TH [32], or orthogonal frequency division multiplexing (OFDM) [33]

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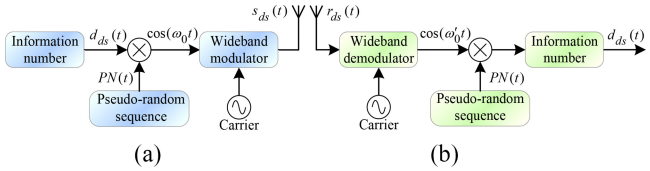


Fig. 2. DS spread spectrum communication system. (a) Transmitter. (b) Receiver.

technologies, DS is better suited for environments with significant interference and multipath effects, as well as scenarios with low power consumption requirements, while also improving signal security. These advantages make DS particularly well-suited for UWB localization systems. Since DS relies on spreading sequences to distinguish signals rather than TH techniques, it does not mitigate multiaccess interference (MAI) [34] through time-domain separation. However, in our experimental setup, only a single tag communicates with multiple anchors for distance estimation at any given time, meaning that MAI does not significantly impact the ranging process. Instead, the primary source of non-Gaussian noise considered in this study arises from NLOS propagation effects.

Accurate distance estimation in UWB wireless positioning networks depends on precise ToF measurements under ideal LOS situation. The most widely used method for this is the double-sided two-way ranging (DS-TWR) technique [5], which is also employed in this article. However, in indoor environments, NLOS situations frequently occur, distorting signal delays, increasing propagation time, and introducing a positive bias in ToF estimation. Many UWB multianchor positioning algorithms assume LOS conditions, leading to significant localization inaccuracies in the presence of NLOS and multipath effects [6]. Thus, detecting and addressing NLOS effects is essential. The CIR represents a channel's response to an impulse, reflecting the characteristics of the communication channel. In UWB systems, each symbol transmitted by a tag or anchor can be expressed as [13]

$$s(t) = \sqrt{A} \sum_{m=0}^{M-1} p\left(t - mT_p\right) \tag{1}$$

where A denotes the amplitude of a transmitted pulse, $p(\cdot)$ represents a single Gaussian pulse waveform, and M pulses with a period of T_p from specific frames, with m being the pulse index within a frame. The CIR $h(t)$ is expressed as

$$h(t) = \sum_{k=1}^{K} \eta_k \delta(t - \alpha_k) \tag{2}$$

where η_k and α_k denote the fading coefficient and time delay for kth path, respectively. K is the total number of paths. Thus, the received signal results from the summation of transmitted signals, each attenuated and delayed along multiple paths, formulated as

$$r(t) = \sum_{k=1}^{K} \eta_k s(t - \alpha_k) + n_z(t). \tag{3}$$

where, $n_z(t)$ denotes the additive white Gaussian noise (AWGN) with variance σ_r^2.

When the DWM1000 receiver receives the preamble from the transmitter, it attempts to align the incoming message with the preamble. Then, it performs a correlation calculation between the incoming signal and the initial guess. The result of this correlation is the CIR observed in the graph, which reflects the arrival times and strengths of the different paths the signal encounters during propagation, such as reflections and refractions. Therefore, the CIR can be used for NLOS detection. The duration of the CIR is independent of the duration of the transmitted signal, as it is primarily determined by the channel's multipath characteristics and signal propagation delay. Due to multipath effects, the initial guess is more likely to be late rather than early, so the DWM1000 aligns the initial guess at approximately 3/4 of the CIR memory, corresponding to a position around 750 ns [35]. Fig. 3(left) displays CIR curves for different NLOS and LOS conditions. It is evident that in the LOS CIR curve, the first path (FP) consistently reaches the curve's peak, while in NLOS conditions, the FP peak typically does not reach the maximum amplitude. Additionally, the normalized peak amplitude is lower under NLOS than LOS. Multipath effects further impact NLOS CIR curves, resulting in multiple local peaks. The number of these peaks increases with the severity of obstacles; for instance, Fig. 3(e) shows a higher number of local peaks due to a metal plate occlusion than with glass in Fig. 3(b). These distinctive characteristics support the use of ML to distinguish LOS from various NLOS conditions.

B. Time–frequency Diagram of CIR

Considering the distinct characteristics of NLOS and LOS CIR data, it is expected that the peak and FP signals will display different spectral profiles in the frequency domain. To emphasize the frequency-domain features of CIR signals while retaining temporal information, a TFDOCIR is created for identifying LOS and NLOS conditions. The CWT was chosen for generating TFDOCIR due to its effectiveness in capturing specific features in CIR signals and images [36]. Based on wavelet analysis theory, the definition of CWT is

$$\mathrm{CWT}(a, \tau) = \int_{-\infty}^{\infty} r(t) \varphi_{a,\tau}(t) dt \tag{4}$$

where, $r(t)$ represents CIR data, and $\varphi_{a,\tau}(t)$ is the fundamental wavelet $\varphi(t)$ transformed by scaling and translation, which can be expressed as

$$\varphi_{a,\tau}(t) = \frac{1}{\sqrt{|a|}} \varphi\left(\frac{t - \tau}{a}\right) \tag{5}$$

where, a represents the scale factor and τ acts as the translation factor. Here, the generalized Morse wavelet is chosen as the fundamental wavelet, with its frequency-domain representation given by

$$\varphi_{\beta,\gamma}(\omega) = 2\left(\frac{e\gamma}{\beta}\right)^{\frac{\beta}{\gamma}} U(\omega) \omega^{\beta} e^{-\omega^{\gamma}} \tag{6}$$

where, $U(\omega)$ represents the Heaviside step function, $2(e\gamma/\beta)^{[\beta/\gamma]}$ serves as a normalization constant, and γ, β are symmetry parameters set to 3 and 60 in this study to achieve the desired time-bandwidth product. The Morse

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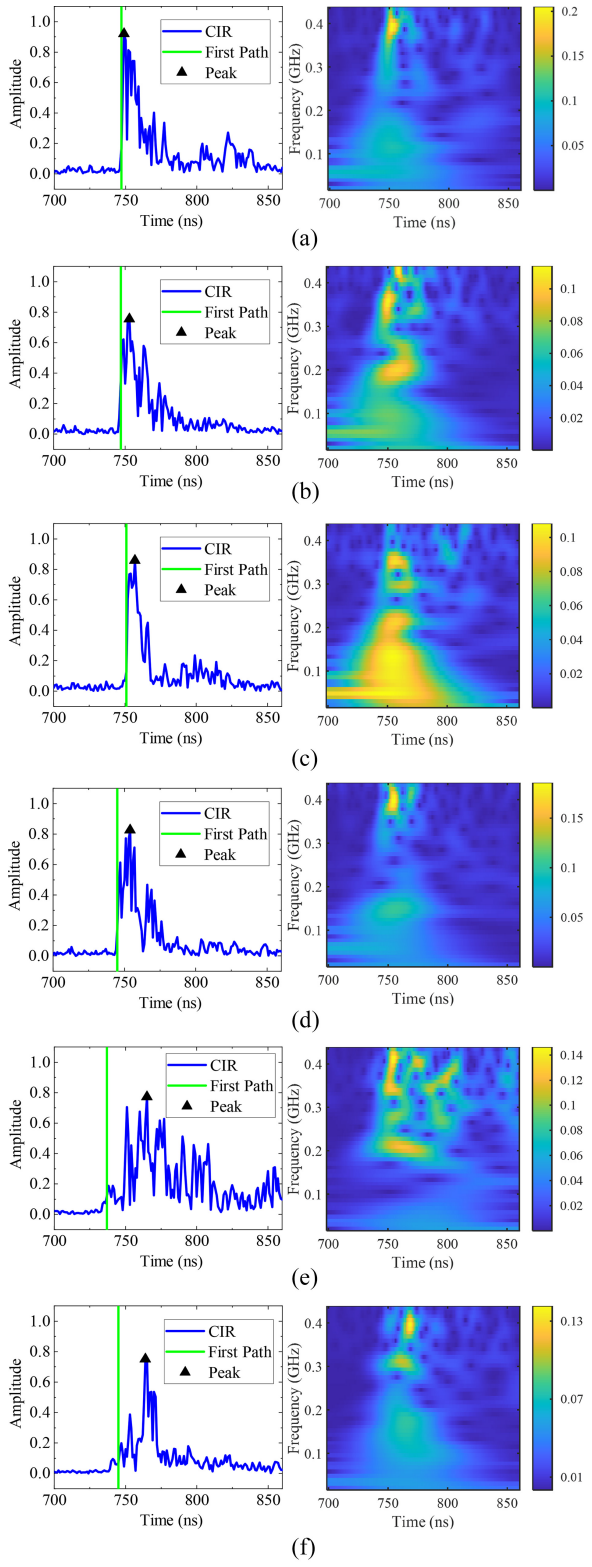


Fig. 3. Comparison of CIR and TFDOCIR under different channel conditions. (a) LOS. (b) Glass. (c) Human. (d) Wooden door. (e) Metal plate. (f) Wall.

wavelet, selected for its zero demodulation skewness and minimal Heisenberg area, is used. Given the CIR data sampled at a 1 ns interval, the scale factor a is set to 128. Finally, the TFDOCIR is generated using the CWT coefficient matrix values from the CIR time-series data. The right side of Fig. 3 presents the TFDOCIRs obtained under different NLOS and LOS scenarios, corresponding one-to-one with the CIR curves on the left. At identical wavelet frequency and time, the energy distribution within the TFDOCIRs varies significantly between NLOS and LOS conditions. Due to multipath delays and attenuation in the CIR signals, the TFDOCIR energy is more concentrated for LOS conditions compared to NLOS. As shown in Fig. 3(e), a higher number of local peaks results in more dispersed energy in the TFDOCIR. Thus, combining the time–frequency characteristics of both CIR and TFDOCIR enhances the feature available for distinguishing between LOS and NLOS conditions.

C. Parallel Spatiotemporal Feature Fusion Network for NLOS Detection

Parallel DL has been effectively applied across domains like image classification and natural language processing. Compared to traditional single-input neural networks, the PSTFFN offers superior performance by training on multiple interrelated features concurrently. In this study, we generate various input vectors, matrices, and manually extracted 1-D features from the same received CIR signal using distinct algorithms. By integrating CIR, TFDOCIR, and manually extracted features, the network harnesses their complementary strengths to substantially improve model performance. The PSTFFN architecture for NLOS identification is illustrated in Fig. 4, with its workflow proceeding as follows.

Step 1: The received CIRs are initially processed through the CWT module to produce TFDOCIRs. Five handcrafted features are then extracted from the CIR. Finally, the CIR, TFDOCIR, and these manually extracted features are each directed to their designated feature extraction modules.

Step 2: The CIR time series feature extraction module consists of three 2-D convolutional layers with different kernel sizes, a BiLSTM layer, and an squeeze-and-excitation (SE) attention mechanism, as shown in Fig. 4(b). After the first two 1-D convolutional layers, batch normalization (BN) layers are applied to normalize the feature maps, reducing the number of training epochs and minimizing generalization error. We use the ReLU activation function and max pooling to accelerate convergence and prevent overfitting. Pooling layers are typically added after convolutional layers to downsample the feature maps, thereby reducing their spatial dimensions.

The deep spatial features obtained from the CNN are passed through the SE attention block, which selectively highlights important channel features. These enhanced features are then input to a BiLSTM to capture spatiotemporal dependencies (last output), and finally, the flattened $\mathbf{X}_{CIR}$ are passed to the feature fusion module. This structure enables the network to manage temporal dependencies and complex sequences efficiently. The SE attention mechanism, shown in Fig. 4(c), first performs global pooling to capture channel statistics, then passes the data through two fully connected (FC) layers to model interdependencies, and finally scales each channel using learned weights to emphasize relevant features [37].

Step 3: To ensure efficient feature extraction, the TFDOCIR 2-D image feature extraction module consists of three 2-D convolutional layers, three max-pooling layers, and the coordinate attention (CA) mechanism. The input TFDOCIR is

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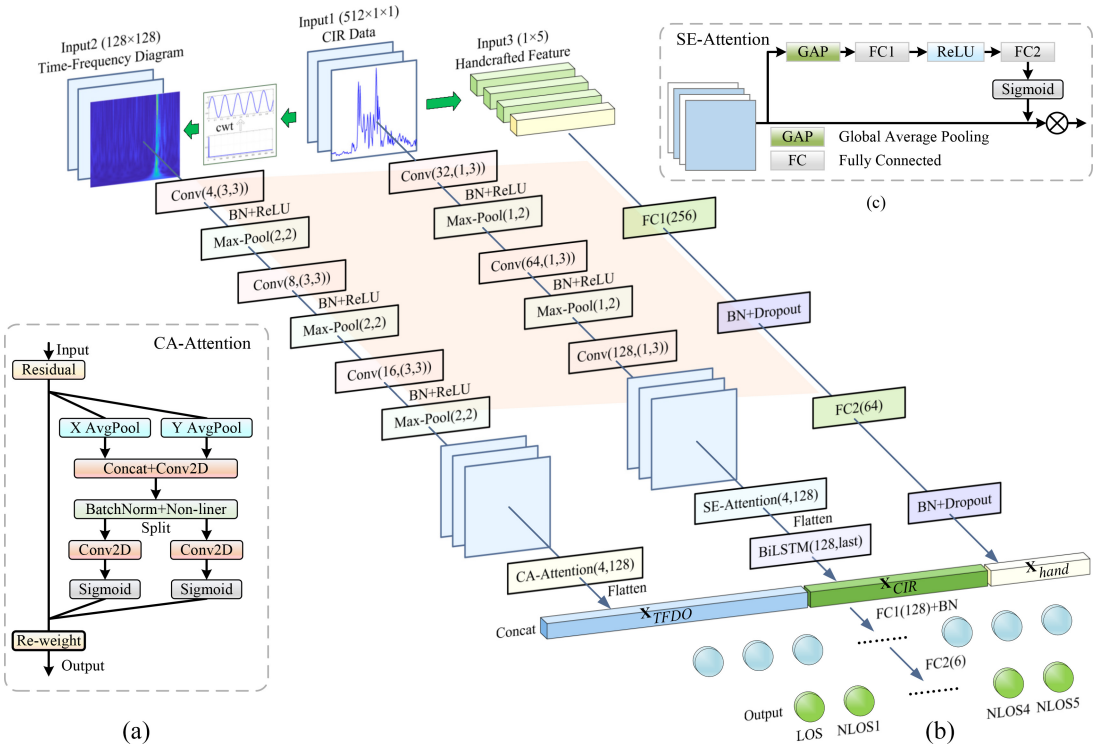


Fig. 4. Network architecture for PSTFFN in NLOS recognition, (a) CA attention block, (b) PSTFFN, and (c) SE attention block.

processed into a 2-D matrix through pixel-wise mathematical transformations. The convolutional layers, initialized with random kernels, perform convolutions on the TFDOCIR input to extract features related to edge. To reduce computational complexity, max-pooling layers are employed to reduce the size of the output matrices from the these layers. After extracting and optimizing the TFDOCIR features, the resulting feature matrix is passed into the CA attention module. The CA attention mechanism enhances the network's ability to handle complex image tasks. The spatial key features extracted by the CA module are then flattened into a 1-D vector. To ensure a balance of information between CIR and TFDOCIR, the output $\mathbf{X}_{TFDO}$ from the flatten layer is passed to the feature fusion layer, where its dimensionality is reduced through a hidden layer. As shown in Fig. 4(a), the CA mechanism splits channel information into horizontal and vertical coordinates, encoding dependencies along each direction. This spatial information is then combined with the input channels to enhance spatial awareness and focus on relevant features [38].

Step 4: The handcrafted feature extraction module comprises two FC layers, each followed by a BN layer and a dropout layer. The FC layers, where FC1 has more hidden units than FC2, are designed to first expand and then compress the handcrafted features, while the BN and dropout layers help mitigate overfitting. The extracted features, denoted as $\mathbf{X}_{hand}$, are then fused with other features in the feature fusion module.

Step 5: The module for feature fusion, as depicted in Fig. 4(b), is the final component of the PSTFFN. It includes a merging layer and a hidden layer, where the 1-D feature vectors from the three extraction modules are combined into a single new vector. This vector is then passed through a hidden layer with a softmax activation, which outputs the class probabilities for classification

$$\text{Softmax}\left(x_{is}^{\text{softmax}}\right) = \frac{\exp\left(x_{is}^{\text{softmax}}\right)}{\sum_{js=1}^{6}\exp\left(x_{js}^{\text{softmax}}\right)} \quad 1 \le js \le 6 \tag{7}$$

where, x_{is}^{softmax} and x_{js}^{softmax} are the isth and jsth input elements, respectively.

Additionally, the model uses focal loss as the multiclass loss function to handle class imbalance by emphasizing hard-to-classify datasets. This loss function incorporates a modulation term to the standard cross-entropy loss, enabling the model to focus more on these challenging instances

$$L = -\alpha_2\left(1-\hat{y}_{iy}\right)^{\gamma}\log\left(\hat{y}_{iy}\right) \tag{8}$$

where, α_2 represents the balancing factor, γ denotes the focusing parameter, $\hat{y}_{iy}$ represents the predicted probability for the iy class.

It is worth noting that the designed PSTFFN network and NLOS recognition process can handle CIR, TFDOCIR, and handcrafted features in parallel. When only one data source (CIR, TFDOCIR, or handcrafted features) is used, the corresponding feature extractor operates, as described in steps 2, 3, or 4 of the NLOS detection workflow. The extracted features are then directly fed into the hidden layer, adjusted to the dimensions needed for NLOS detection in step 5.

The handcrafted characteristics used are as follows.

1) The received first-path signal strength difference (RFPSD) represents an interpolation between the initial

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strength indication and the received signal. The FP signal strength (FPSS) is calculated as follows:

$$\mathrm{FPSS} = 10 \times \lg\left(\frac{F_1^2 + F_2^2 + F_3^2}{\mathrm{PAC}^2}\right) - A \ \mathrm{dBm} \tag{9}$$

where, F_1, F_2, and F_3 denote the first, second, and third harmonics of the FP CIR amplitudes, respectively. The constant A equals 121.74 when the pulse repetition frequency (PRF) is 64 MHz as noted in [5], and PAC represents the preamble accumulation count.

Obtain the final RFPSD

$$\mathrm{RFPSD} = \mathrm{FPSS} - \mathrm{RSSI} \ \mathrm{dBm} \tag{10}$$

where, RSSI is the received signal strength indication [39].

2) Ranging distance error (RDE) denotes the discrepancy between the ToF-derived range and the estimated NLOS FP distance, calculated as follows:

$$d_1 = \frac{10^{\frac{P_T + G_R + 20\lg(c) - \mathrm{FPSS} - 20\lg(4\pi f_R)}{20}}}{200} \tag{11}$$

where, P_T represents the transmission power at 33.5 dB/TX Gain, G_R signifies the total antenna gain of both the anchor and tag antennas, each at 3 dBi, c is the propagation speed of UWB, and f_R denotes the center frequency at 6489.6MHz.

The RDE is obtained as follows:

$$\mathrm{RDE} = |d_2 - d_1| \tag{12}$$

where, d_2 is the ranging result from the UWB ToF method.

For the additional handcrafted features, we used the kurtosis factor k [16], mean excess delay τ_{med} [17], and root-mean-square delay spread τ_{rms} [17].

D. Ranging Error Mitigation Method

We assume that the actual distance between the tag and the anchor is d^*. The estimated distance can then be expressed as

$$\begin{cases} d_{\mathrm{LOS}} = d^* + \eta(t) \\ d_{\mathrm{NLOS}} = d^* + \Delta d(t) + \eta(t) \end{cases} \tag{13}$$

where, d_{LOS} and d_{NLOS} represent the UWB measurements for NLOS and LOS conditions, respectively. $\Delta d(t)$ represents the extra error caused by NLOS, which is typically non-Gaussian, $\eta(t)$ denotes the UWB measurement noise.

Following NLOS detection with the PSTFFN and obtaining classification outcomes, the dataset is divided into more homogeneous subclasses. This reduces intraclass noise heterogeneity, allowing the prediction model to focus on subclass-specific patterns, such as certain types of non-Gaussian error distributions, and apply targeted processing accordingly. Subsequently, the STARN is applied to predict UWB ToF ranging errors for different subclass datasets, represented as $\Delta d(t)$ and $\eta(t)$. These error values are then subtracted from the ToF measurements to reduce ranging inaccuracies, as shown by subtracting the right side of (13) from the left. The corrected distance estimates are subsequently used to calculate the target's relative position using the LSs [5] positioning algorithm.

Considering computational limitations and efficiency, the proposed STARN for mitigating UWB ranging errors employs a dual-input structure with multiple convolutional layers followed by a BiLSTM, as shown in Fig. 5. First, we extract the tail data of the raw UWB CIR (512 × 1) and perform tensor reconstruction (512 × 1 × 1) as Input 1. Additionally, seven handcrafted features (ToF ranging value, RFPSD, RSSI, RDE, kurtosis factor k [16], mean excess delay τ_{med} [17], and root-mean-square delay spread τ_{rms} [17]) are used as auxiliary Input 2 to enhance the model's predictive accuracy and generalization [40].

The CIR data is first processed through a multilayer deep convolutional feature extraction network. Each of the initial convolutional layers is trailed by a BN layer and a max-pooling operation for downsampling. After the last convolutional layer, an SE attention mechanism is applied to emphasize key channel features, indirectly enhancing the model's robustness to non-Gaussian noise. The resulting features are then flattened and fed into a BiLSTM layer (using only the last time step's output) to capture spatiotemporal dependencies, ultimately directing the output to the feature fusion layer.

Meanwhile, the manually extracted features (1x7) are processed through three FC layers to capture nonlinear relationships in the data before being sent to the feature fusion layer. A dropout layer is added between FC1 and FC2 to prevent overfitting, given the limited number of learnable parameters in the model. A sigmoid activation is applied at the end to normalize the feature values to the range [0,1]. In the fusion module, the high-level features from the CIR data are concatenated and integrated with the enhanced manually extracted features into a new vector. This new feature vector then passes through two FC layers (FC2 as output), gradually reducing the feature dimension and smoothing the representation before entering the regression layer for the final prediction.

Given the significant range of UWB ranging errors, particularly when severe obstructions cause errors to exceed 1 m, MSE is used as the loss function for the ranging error predictor. Unlike mean absolute error (MAE), MSE gives greater weight to larger errors, helping to reduce them and improve the model's overall performance. MSE quantifies the average of the squared differences between the actual and predicted ranging errors, reflecting the deviation from the true values. The formula for calculating MSE is provided below

$$\mathrm{MSE} = \frac{1}{n}\sum_{i=1}^{n}\left(\Delta d_i' - \Delta d_i\right)^2. \tag{14}$$

where, $\Delta d_i'$ represents the estimated ranging error, Δd_i denotes the true ranging error, n stands for the number of samples.

The workflow for the proposed PSTFFN+STARN approach is illustrated in Fig. 6. First, online CIR data, TFDOCIR obtained through CWT transformation, and manually extracted features are input into the trained PSTFFN classification model for NLOS detection, yielding LOS or one of five specific NLOS propagation scenarios. If PSTFFN identifies the input data as LOS, it bypasses ranging error mitigation and directly contributes to the positioning calculation. Otherwise, the appropriate NLOS ranging error prediction model is selected based on the detected NLOS type for error mitigation. The mitigated NLOS ranging values and LOS ranging values are

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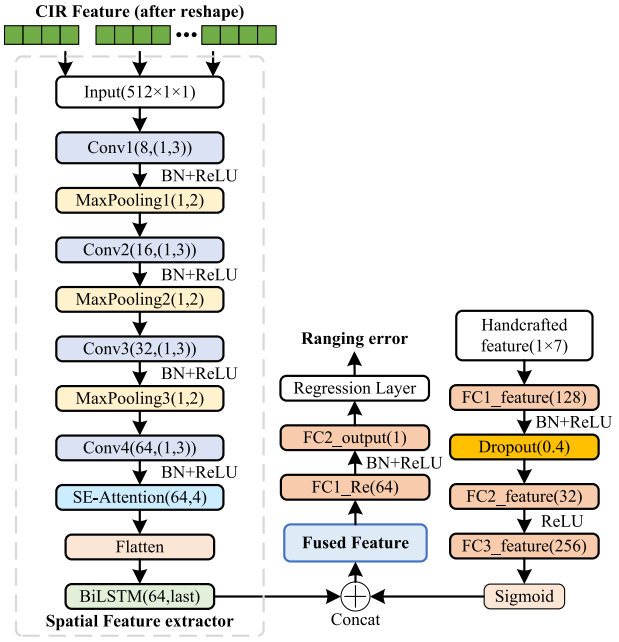


Fig. 5. Diagram of the STARN structure for NLOS error prediction.

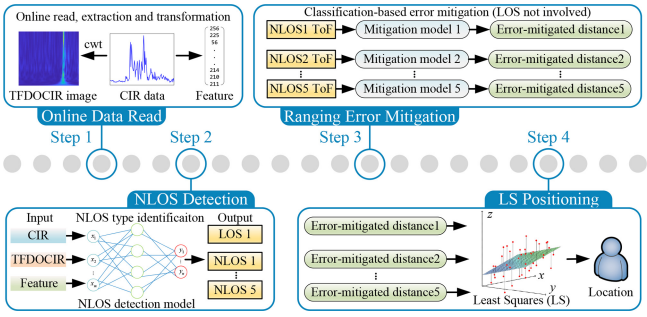


Fig. 6. Workflow of online NLOS detection and error mitigation.

then jointly processed in an LS calculation to obtain the final mitigated positioning result.

IV. Experimental Setup and Verification

This section presents an evaluation of our proposed NLOS binary and multiclass recognition, using comprehensive simulations and CIR signals collected from two different types of positioning scenarios. Next, we demonstrate the effectiveness of our ranging error mitigation method by using the collected ranging values, and the measured ground truth. Finally, we assess the performance of the proposed approach through positioning tests conducted in real-world scenarios.

A. Experimental Setup

We evaluated performance of our proposed model for NLOS detection in a binary classification task using a publicly available dataset from [41]. The data was gathered from seven distinct environments: Office 1, Office 2, a small apartment, a workshop, a kitchen-living room, a bedroom, and a boiler room. Each scene contributed 3000 LOS samples and 3000 NLOS samples, bringing the total to 42 000 samples.

As detailed multiclass CIR datasets are not readily available online, we gathered experimental data from a building

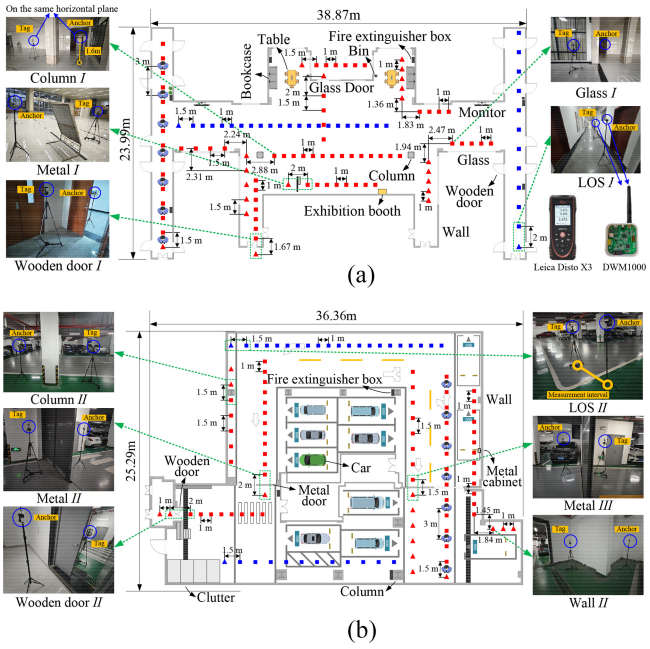


Fig. 7. Experimental setup and samples collection details. The anchor ("Δ") and tag ("□") are placed as illustrated. LOS ranging positions are marked in blue, while those under NLOS conditions are depicted in red. (a) Hallway. (b) Underground parking garage.

lobby and an underground parking garage, as illustrated in Fig. 7. The dataset includes five common types of indoor occlusions: glass, human bodies, wooden doors, metal plates, and walls, with 4000 samples for each type. We gathered and filtered a total of 24 000 samples across six classes, including NLOS occlusions and LOS CIR signals, for model training and testing. The data was collected using the DecaWave DWM1000 module, mounted on a tripod at 1.6 m. Since higher center frequencies in UWB systems facilitate the observation of step changes in the output for diagnosis [39], channel 5 (6489.6 MHz) was selected for the experiment to ensure reliable distance measurement in the indoor environment considered in this study, which includes significant occlusions with a maximum range of 42 m. Additionally, to enhance robustness against interference and mitigate the effects of multipath propagation and signal attenuation, a low data rate of 110 Kb/s was employed, with a preamble length of 1024. Both the transmitter and receiver were equipped with 3 dBi omnidirectional antennas. The other setup details are provided in Table I. All measurements followed the UWB IEEE 802.15.4a standard, and the distance between base stations and tags was recorded with a Leica Disto X3 laser device, accurate to ±1 mm, as illustrated in Fig. 7(a).

1) *Hallway:* The environment is a hallway located on the second floor of a university laboratory building, containing obstacles, such as a human I (1.68 m, 53 kg), glass windows (0.012 m), glass door (0.02 m), wooden door I (0.03 m), metal plate (0.08 m), tiled walls, and glass curtain walls. The corridor also has metal fireproof cabinets, desks, bookshelves, and glass doors. Fig. 7(a) marks the positions of the anchor "Δ" and the tag "□" during data collection. Throughout the collection of each batch of CIR data, the anchor stays fixed in place, whereas the tag is relocated by a set distance to

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TABLE I
DWM1000 SETTINGS

Parameter	Set value
Channel	5
Central frequency	6489.6 MHz
Bandwidth	499.2 MHz
Pulse repetition frequency (PRF)	64 MHz
Inter-ranging delay	200 ms

gather the data. For the next batch, the anchor is relocated. The distance between person I and the tag behind them remains fixed at 0.75 m, and both person I and the anchor stay in the same position throughout the data collection. After each set of data is gathered, the tag is moved, and after completing one batch, the anchor is relocated by 3 m.

2) *Underground Parking Garage:* Unlike the hallway, the underground garage beneath the office features larger and thicker metal plates (0.12 m), resulting in more severe radio signal attenuation and reflection, which affect signal transmission and coverage. Additionally, reinforced concrete walls separate each parking area in the garage. In this environment, CIR data were collected for NLOS obstructions caused by a wooden door II (0.08 m), human II (1.8 m, 70 kg), metal plate II (0.12 m), and concrete walls, as well as LOS II. The data collection and range measurement methods between the tag and anchor were the same as in the hallway, with the specific anchor and tag placement shown in Fig. 7(b).

This study evaluated the model using four commonly employed metrics: precision, recall, accuracy, and F1 score to assess the classification performance [24]. The prediction results are evaluated using the MAE and root-mean-square error (RMSE), where smaller values of these metrics indicate higher prediction accuracy. Additionally, the mitigation rate is used to assess the effectiveness of error mitigation. The two metrics are defined as follows:

$$\mathrm{MAE} = \frac{1}{n}\sum_{n=1}^{n}\left(\left|d_i' - d_i\right|\right) \tag{15}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{n}\sum_{n=1}^{n}\left(\left|d_i' - d_i\right|\right)^2} \tag{16}$$

where, d_i' represents the true value and d_i denotes the measured value of ranging, n represents the number of inputs. To calculate the MAE and RMSE for 2-D coordinates, d_i' and d_i should be replaced with x_i' and x_i, respectively.

Signals processing and model training were conducted on a computer equipped with an I5-12600KF processor, an RTX 2060S graphics card, and 32GB of RAM. We employ a grid search method to determine the optimal network hyperparameter combinations and layer configurations for the PSTFFN and STARN models. To reduce computational cost, we first constrain the search space for hyperparameters and layer numbers within each module as shown in Table II. The selection of these layer ranges is based on a balance between computational efficiency and model performance. A smaller range (e.g., [2,3,4] for TFDOCIR) helps control computational complexity. Meanwhile, the spatial feature extractor in the STARN adopts a broader search range [2,3,4,5] to enhance spatial feature extraction capabilities. In contrast, the handcrafted feature extractor and classification/regression head require fewer FC layers for effective feature mapping and task completion, thus their search ranges remain narrower.

In the grid search process, given that the PSTFFN and STARN models contain multiple branches and hierarchical structures, we first fix the parameters of other branches and modules, including the batch size and learning rate used for model training. We then perform a localized search on the target module to determine its optimal layer configuration and hyperparameter combination. Once the best configurations for each branch are obtained, we proceed with a global optimization to identify the optimal hyperparameter settings for different branch combinations. Additionally, to prevent overfitting, we adopt K-fold cross-validation (K=5) during hyperparameter selection. The optimal network hyperparameter configurations and layer selections obtained through this process are presented in Figs. 4 and 5.

Upon determining the network architecture, we further optimize the training hyperparameters to enhance convergence speed and stability. For PSTFFN, the optimal batch size is found to be 32, with an initial learning rate of 0.0001. For STARN, the optimal batch size is 64, with an initial learning rate of 0.001. Both models adopt Adam as the optimizer. Finally, the training epochs are determined by initially setting it to 100 for PSTFFN and 200 for STARN, then identifying the optimal values based on the training and validation loss curves, which converge at 20 and 40 epochs, respectively. To further enhance model generalization and mitigate overfitting, L2 regularization, dropout, and learning rate decay are applied throughout the training process of PSTFFN. In contrast, for STARN, only L2 regularization is utilized to prevent excessive weight constraints while improving model stability.

B. Results and Analysis

This article examines the proposed PSTFFN and STARN combined approach from four perspectives. First, we assess the effectiveness of PSTFFN on the traditional binary classification task of NLOS/LOS detection. Next, we assess its performance in distinguishing between LOS and various NLOS scenarios to further test the approach's capability in multiclass classification. Third, ablation experiments are performed to verify the impact of the proposed "detection-then-mitigation" strategy, demonstrating its superiority over direct error regression mitigation methods. Finally, we test the proposed approach in an actual indoor positioning experiment, confirming its enhancement of positioning system accuracy and robustness.

1) Comparative Evaluation of NLOS Binary Classification Models: First, we evaluate the PSTFFN model's performance in conventional binary classification and compare it to four ML (including DL) approaches: SVM [16], EM-GMM [17], MWT-CNN [18], and FCN-Attention [19], as well as with the baseline method PSTFFN-W, which excludes the five handcrafted features. This baseline was included to assess

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TABLE II
HYPERPARAMETER SELECTION FOR PSTFFN AND STARN MODELS

Model	Module	Hyper-parameter	Parameter Selection
PSTFFN	CIR Time Series Feature Extractor	Conv & Pool Layers	{2,3,4}
		Conv Filters	{32,64,128}
		Conv Kernel	{(1,3),(3,3),(5,5)}
		MaxPooling Size	{(2,2),(1,2)}
		SE Reduction Ratio	{4,8,16,32}
		SE Num Channels	{64,128,256}
		BiLSTM Hidden Units	{64,128,256,512}
	TFDOCIR 2D Image Feature Extractor	Conv & Pool Layers	{2,3,4}
		Conv Filters	{4,8,16,32}
		Conv Kernel	{(1,3),(3,3),(5,5)}
		MaxPooling Size	{(2,2),(3,3)}
		CA Reduction Ratio	{4,8,16,32}
		CA Num Channels	{64,128,256}
	Handcrafted Feature Extractor	FC Layers	{2,3}
		Units in FC	{32,64,128,256}
		Dropout Rate	{0.2,0.3,0.5}
	Classification Head	FC Layers	{2,3}
		Units in FC	{64,128,256}
	Training Hyper-parameters	Batch Size	{16,32,64}
		Learning Rate	{0.001,0.0001,0.00001}
		Optimizer	{Adam,SGD,RMSprop}
STARN	Spatial Feature Extractor	Conv & Pool Layers	{2,3,4,5}
		Conv Filters	{8,16,32,64,128}
		Conv Kernel	{(1,3),(3,3),(5,5)}
		MaxPooling Size	{(2,2),(1,2)}
		SE Reduction Ratio	{4,8,16,32}
		SE Num Channels	{64,128,256}
		BiLSTM Hidden Units	{64,128,256,512}
	Handcrafted Feature Extractor	FC Layers	{2,3,4}
		Units in FC	{16,32,64,128,256}
		Dropout Rate	{0.2,0.3,0.5}
	Regression Head	FC Layers	{2,3}
		Units in FC	{16,32,64,128,256}
	Training Hyper-parameters	Batch Size	{32,64,128}
		Learning Rate	{0.001,0.0001,0.00001}
		Optimizer	{Adam,SGD,RMSprop}

the impact of incorporating handcrafted features on model performance. All methods utilize the original parameters and model structures from [16], [17], [18], and [19], with adjustments made to the training and test set configurations. The training data was drawn from the four circumstances of dataset [41]: a kitchen with a living room, office 1, a small workshop, and a small apartment, each contributing 1/4 of the data. This resulted in 20 000 CIR samples, evenly divided between NLOS and LOS. Evaluation was performed on three remaining scenarios: office 2, a boiler room and a bedroom (each accounting for 1/3 of the data), making up 12 000 CIR data sets, also evenly divided between NLOS and LOS. This arrangement ensured a clear distinction between training and testing sets, validating the model's generalization ability.

In the comparison methods, SVM utilizes manually extracted CIR features, including energy ε_r, rise time t_{rise}, maximum amplitude r_{max}, root mean square delay spread τ_{rms}, mean excess delay τ_{med}, estimated distance d_t, and kurtosis k as input for training. EM-GMM [17] uses only τ_{med}, τ_{rms}, and the number of paths N_P that account for over 85% of the total received signal energy. MWT-CNN [18] employs Morlet wavelet-transformed CIR signals, resulting in a 128 × 128 wavelet power spectral image as input. FCN-Attention [19] uses normalized CIR signal data, while PSTFFN-W incorporates both CIR data and TFDOCIR images (128 × 128 × 3) derived from CWT-transformed CIR data, as outlined in Table III. The calculation approaches for the handcrafted CIR features are detailed in [16] and [17]. To ensure fairness, the same hyperparameters and training sets for SVM, EM-GMM, MWT-CNN, and FCN-Attention were set according to [16], [17], [18], and [19], respectively. PSTFFN-W and PSTFFN used the hyperparameters and training configurations specified in Section IV-A.

TABLE III
FEATURE AND DATA UTILIZATION IN NLOS BINARY CLASSIFICATION TECHNIQUES

Method	**Data or feature employed**
SVM [16]	CIR feature ε_r, t_{rise}, r_{max}, τ_{rms}, τ_{med}, d_t, k
EM-GMM [17]	CIR feature τ_{med}, τ_{rms} and N_P
MWT-CNN [18]	CIR image
FCN-Attention [19]	CIR signal $r(t)$
PSTFFN-W	CIR signal $r(t)$ and TFDOCIR
PSTFFN	CIR signal $r(t)$, TFDOCIR, CIR feature τ_{med}, τ_{rms}, k, RDE and RFPSD

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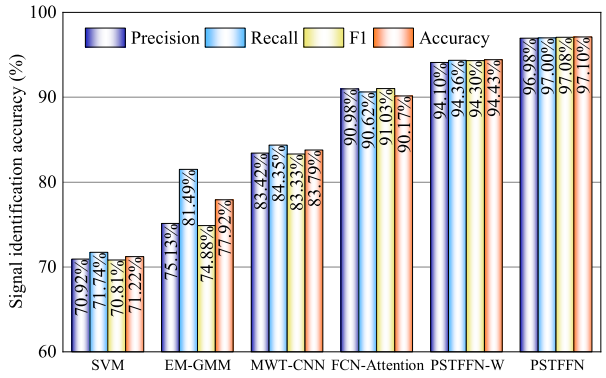


Fig. 8. Evaluating the performance of various NLOS binary classification methods.

Fig. 8 compares the performance of various methods in NLOS/LOS binary classification, using four evaluation metrics for each ML method. The results show that, even in cross-environment NLOS binary classification (where the test and training data come from different environments, with the test data includes three distinct situations), the proposed PSTFFN approach exhibits better generalization than the other approaches. Both PSTFFN and the baseline PSTFFN-W (which lacks manually crafted features) achieved over 94% across all metrics, surpassing conventional approaches in distinguishing NLOS and LOS situations. This advantage is attributed to the use of TFDOCIR image data, which captures richer spatial and temporal information about signal behavior under various conditions. TFDOCIR, in particular, visualizes the signal's energy distribution over time and frequency, offering more detailed insights than raw CIR data or CIR features alone.

Compared to PSTFFN-W, PSTFFN improves accuracy, recall, precision, and F1 score by 2.67%, 2.78%, 2.64%, and 2.88%, respectively, demonstrating that the inclusion of five handcrafted features (τ_{med}, τ_{rms}, k, RDE, and RFPSD) significantly enhances the model's performance in NLOS binary classification task. Among the remaining four methods, FCN-Attention records the highest accuracy at 90.17%, followed by MWT-CNN at 83.79%, and EM-GMM at 77.92%. Despite EM-GMM employing an unsupervised learning approach that achieves relatively high recall, its overall performance is unsatisfactory. The SVM-based method works the worst, achieving only 71.22%. This poor performance is likely due to SVM's sensitivity to the training data distribution. The substantial differences between the test set scenarios (office, bedroom, and boiler room) and the training set hinder SVM's ability to generalize effectively, as it struggles to capture complex NLOS features across diverse environments. Additionally, the handcrafted features show significant variability across different environments, further limiting SVM's performance.

To evaluate the impact of integrating SE and CA attention mechanisms into the feature extractor, ablation experiments were performed with the following comparisons: 1) PSTFFN without attention, where both flatten layers, bypassing the attention mechanisms and connecting directly to the maxpool layer and Conv(128, (1,3)) layer; 2) PSTFFN with CA, where only the Flatten layer in the CIR feature extraction part is connected directly to the Conv(128, (1,3)) layer; 3) PSTFFN with CA, where only the flatten layer in the TFDOCIR image feature extraction part is connected directly to the maxpool layer, without any other structural changes; and 4) PSTFFN with both CA and SE mechanisms. The structures of each feature extractor are presented in Table IV. For consistency, the SE and CA attention modules are set with reductionRatio=4 and numChannels=128, utilizing the same dataset and hyperparameters as in Table II and the binary classification experiment discussed earlier.

Table IV summarizes the identification accuracy of each model on the NLOS binary classification test data, along with the corresponding number of trainable parameters, total training time, and the average classification time per sample (computed over 100 test runs, including data processing time, highlighted in red). As shown, our proposed PSTFFN with all delivers the best overall identification performance. Although the model's parameter count and training time have increased (50.5 M and 911.41 s, respectively), its accuracy improved by 4.67% compared to the baseline PSTFFN without attention. In comparison, PSTFFN with SE and PSTFFN with CA also perform antly well, achieving classification accuracies of 94.99% and 95.87%, respectively. Models with fewer trainable parameters, such as PSTFFN with SE, require less training time (829.65 s) and achieve faster classification response (0.279 ms). Each attention mechanism has distinct strengths: the CA mechanism enhances feature emphasis, particularly beneficial for image recognition tasks, while the SE mechanism boosts detection accuracy by effectively focusing on key CIR features. The superior performance of the proposed model suggests that combining both mechanisms optimizes feature extraction for CIR and TFDOCIR, ultimately improving overall accuracy. Notably, this enhancement adds minimal overhead, with nearly unchanged parameters (49.6 M) compared to the baseline without attention. Furthermore, the model performs NLOS classification in just 1.339 ms, with data preprocessing being the primary contributor to the total execution time. Based on 100 trials, operations, such as manual feature computation, TFDOCIR data processing, and tensor reconstruction require an average of 0.974 ms. This efficiency ensures the model's capability to support real-time localization tasks.

2) Evaluation of NLOS Multiclass Classification Performance: To further validate effectiveness of our approach in distinguishing different NLOS scenarios, we compared it against LSTM [15], CNN+LSTM [23], CNN-A [14], MIL [25], and the baseline method PSTFFN-W, which does not use any handcrafted feature enhancement. The comparison methods, MIL, CNN-A, CNN+LSTM, and LSTM, utilize their respective original parameters and model structures from [14], [23], [25], and [15].

The multiclass model's training and testing data were gathered from the hallway and underground garage scenes in Fig. 8(a) and (b), as described in Section IV-A. We used 18 000 CIR samples from the scene shown in Fig. 8(b), including a wooden door II (0.08 m), a human II (1.8 m, 70 kg), metal plates II/III (0.12 m), and concrete walls, as well as from

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TABLE IV
VARIATIONS IN FEATURE EXTRACTION ACROSS MODELS

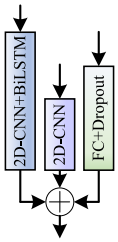
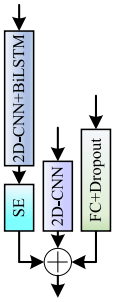
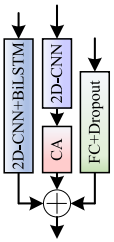
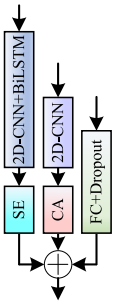
Model	PSTFFN without attention	PSTFFN with SE	PSTFFN with CA	PSTFFN with all
Structure				
Parameters(M)	49.3	49.6	50.2	50.5
Accuracy(%)	92.43	94.99	95.87	97.10
Training time(s)	818.73	829.65	852.07	911.41
Classification time(ms)	0.262+0.974	0.279+0.974	0.344+0.974	0.365+0.974

TABLE V
FEATURE AND DATA UTILIZATION IN NLOS MULTICLASSIFICATION TECHNIQUES

Method	Feature or data used
LSTM [15]	CIR signal $r(t)$
CNN+LSTM [23]	CIR signal $r(t)$
CNN-A [14]	CIR coefficients image
MIL [25]	CIR signal $r(t)$ and TFDOCIR
PSTFFN-W	CIR signal $r(t)$ and TFDOCIR
PSTFFN	CIR signal $r(t)$, TFDOCIR, CIR feature τ_{med}, τ_{rms}, k, RDE and RFPSD

TABLE VI
ASSESSMENT OF VARIOUS TECHNIQUES IN NLOS MULTICLASS CLASSIFICATION

Method	Accuracy/%	Precision/%	Recall/%	F1/%
LSTM	71.84	73.90	69.53	70.51
CNN+LSTM	83.95	84.57	83.26	83.57
CNN-A	83.64	84.12	79.03	79.25
MIL	90.37	90.97	90.13	90.43
PSTFFN-W	93.86	94.18	93.69	93.85
PSTFFN	96.74	96.29	96.37	96.15

the glass doors (0.2 m) in Fig. 8(a), with each class accounting for 1/6 of the total. For evaluation, we tested the model using 7884 CIR samples from the hallway scene in Fig. 8(a), which included a human (1.68 m, 53 kg), glass doors and windows (0.012 m), a wooden door I (0.03 m), metal plates (0.08 m), tiled walls, and glass curtain walls. This separation of training and testing sets ensures a proper assessment of the model's performance in multiclass classification.

In the multiclassification experiments, both LSTM [15] and CNN+LSTM [23] use normalized CIR. CNN-A utilizes the wavelet envelope decomposition method in [14] to generate a 128×128 coefficients image from the CIR. Among the other comparison methods, MIL and PSTFFN-W use normalized CIR and TFDOCIR images derived from CWT-transformed data, as shown in Table V. For a fair comparison, the hyperparameters and training configuration of the LSTM and CNN+LSTM follow those in [15] and [23]. Since [14] and [25] do not specify training parameters for CNN-A and MIL, we adopted the same settings used for CNN-A, MIL, PSTFFN-W, and PSTFFN follow the settings from Section IV-A.

Table VI compares the NLOS multiclass identification performance of various methods. Among them, the LSTM achieves the lowest accuracy, mainly due to its limited ability to extract complex spatial features from CIR for distinguishing different NLOS conditions. Both CNN+LSTM and CNN-A achieve comparable performance. The former achieves slightly higher accuracy (83.95%) owing to its deep spatiotemporal feature extraction, while the latter, using CIR coefficients images, attains 83.64%, possibly constrained by the CNN's feature extraction capacity. In contrast, PSTFFN, PSTFFN-W, and MIL employ multifeature input network architectures, thereby achieving superior multiclass performance over the aforementioned single-input approaches. Notably, PSTFFN further improves upon PSTFFN-W, with increases of 2.88% in accuracy, 2.68% in recall, 2.11% in precision, and 2.3% in F1. These results demonstrate that incorporating five handcrafted features improves the model's ability to distinguish scenarios with similar CIR characteristics, thereby enhancing PSTFFN's accuracy and generalization in multiclass tasks across different environments. Additionally, in multiclass tasks, the number of learnable parameters in PSTFFN remains nearly unchanged compared to binary classification, and the classification time is consistent with Table IV.

The confusion matrix offers additional insights into how different approaches perform in the multiclass classification task. As shown in Fig. 9(a) to (d), Sec.1 to Sec.6 represent LOS, glass, human, wooden door, metal plate, and wall occlusion scenarios, respectively. Among the methods compared, PSTFFN-W and MIL demonstrated some capability for multiscenario recognition. The addition of five handcrafted features notably enhanced PSTFFN's accuracy in Sec.1, Sec.2, and Sec.4. Fig. 9(d) and (f) highlight that both MIL and PSTFFN-W utilized CIR and TFDOCIR images as inputs, employing similar CNN and RNN architectures. While MIL outperformed PSTFFN in Sec.6, its performance in Sec.1, Sec.2, and Sec.4 was slightly lower due to errors in these

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sections, leading to a marginally lower overall accuracy than PSTFFN and PSTFFN-W. In contrast, PSTFFN-W's SE and CA modules enhance critical CIR and TFDOCIR features, yielding improved performance compared to MIL.

3) Evaluation of NLOS Range Error Mitigation Performance: To assess the NLOS mitigation capabilities of the proposed approach and examine the benefits of the strategy that first conducts NLOS detection followed by corresponding range error mitigation techniques, we compared the combined error mitigation method of PSTFFN+STARN with a direct error prediction mitigation approach using STARN, as well as unprocessed UWB ToF errors. In this context, we used 18 000 CIR data samples from the Hallway [Fig. 7(a)] to train the model, which included scenarios, such as a human I, glass doors and windows (0.012 m), wooden door I (0.03 m), metal plate (0.08 m), tiled walls, and glass curtain walls, with each type (6 major categories) representing 1/6 of the data.

In contrast to the configurations used in the NLOS identification experiments, the test data consists of 8000 CIR samples from the underground parking garage shown in Fig. 7(b), which includes a wooden door II (0.08 m), human II (1.8 m, 70 kg), metal plates II/III (0.12 m), concrete walls, and the glass door (0.2 m, shown in Fig. 7(a)). Additionally, the training approach for directly using STARN for error mitigation differs from the PSTFFN+STARN combination. The latter trains six different STARN models based on range error data from six distinct scene categories, while the former does not distinguish between NLOS conditions and only trains a single STARN model. Sec.1 to 6 represent LOS, glass, human, wooden door, metal plate, and wall, in that order.

Fig. 10 presents histograms of UWB ranging error for the PSTFFN+STARN combined approach under five NLOS conditions and one LOS condition, including original, predicted, and mitigated errors. It can be observed that the proposed approach effectively alleviates the non-Gaussian ranging errors across various NLOS scenarios and suppresses LOS measurement noise to some extent. This improvement stems from PSTFFN's fine-grained NLOS classification, which partitions data into more homogeneous subclasses, thereby isolating the influence of certain noise components. As a result, the original error distributions in Fig. 10 (a)–(d) exhibit a closer resemblance to a Gaussian distribution after PSTFFN. By integrating SE attention and manually extracted features, STARN enhances sensitivity to key features and improves robustness to noise. As a result, the predicted error distribution (blue) nearly overlaps with the original error (red), appearing purple in overlapping regions. Moreover, the method partially mitigates negative ranging errors caused by early arriving multipath components.

Table VII presents the RMSE and MAE of range mitigation using the PSTFFN+STARN combination method across five distinct NLOS scenarios and the LOS scenario, compared to the original errors. Additionally, the table provides the computation time for STARN-based error mitigation (excluding PSTFFN preclassification). All reported results are averaged over 100 error mitigation trials per scenario. It is evident that after applying the PSTFFN+STARN combination method, the RMSE and MAE for each NLOS situation in Table VII are significantly reduced, with overall MAE and RMSE decreasing

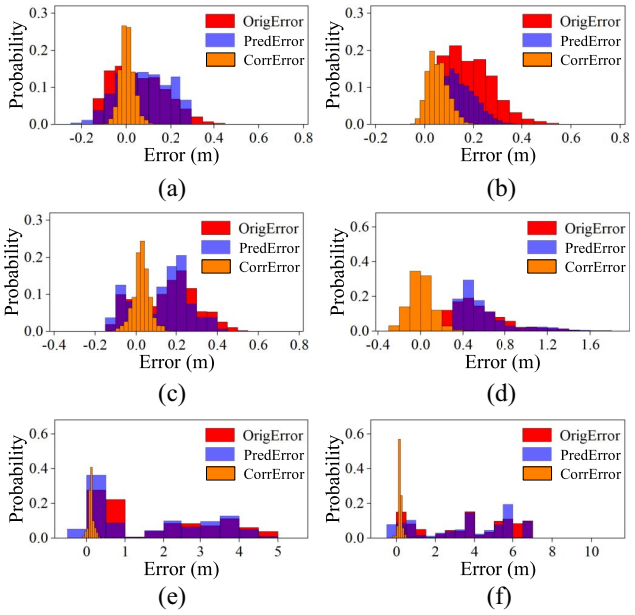


Fig. 10. Comparison of error histograms under different occlusion conditions: raw errors (red), predicted errors (blue), and corrected errors (orange), (a) Sec.1, (b) Sec.2, (c) Sec.3, (d) Sec.4, (e) Sec.5, and (f) Sec.6.

by 92.72% and 95.51%, respectively. Additionally, the error mitigation rates between Sec.2 and Sec.3 are quite similar, with Sec.3 exhibiting a RMSE mitigation rate of 75.03%, which is lower than Sec.2's 76.45%. This result is unexpected, especially considering that Sec.3 had a higher original RMSE. Combined with the multiclass results from the confusion matrix, it is likely that PSTFFN misidentified instances in Sec.3, resulting in insufficient mitigation of the range errors. Compared to PSTFFN, STARN has a lower computational cost, with only 2.2M trainable parameters and an average training time of just 8.382 s. Across experiments in six types of scenarios, it completes distance error prediction and mitigation for a single sample in an average of 0.441 ms (including 0.247 ms for data preprocessing), which is only 33% of the time required by PSTFFN, enabling a faster real-time response.

Fig. 11 compares the 2-D cumulative distribution functions (CDFs) of ranging errors for the original ToF errors, the PSTFFN+STARN error mitigation method, and the direct STARN error prediction mitigation method. As shown, the approach that uses PSTFFN for NLOS identification followed by STARN for error prediction mitigation (green) effectively suppresses NLOS errors, achieving a mitigation rate of over 46% in 95% of NLOS cases compared to the original error (red). This combined method also demonstrates greater robustness than using direct STARN-based error prediction (blue), as it better reduces ranging errors. Particularly in Sec.5 and Sec.6, representing metal plate and concrete wall scenarios, the 95% error mitigation rate reached 91.30% and 91.24%, respectively, as illustrated in Fig. 11(e) and (f).

In contrast, the direct STARN prediction method overlaps almost entirely with the original error (red curve) in the LOS, glass, and human occlusion scenarios shown in Fig. 11(a)–(c), offering minimal error mitigation. Notably, in the Glass

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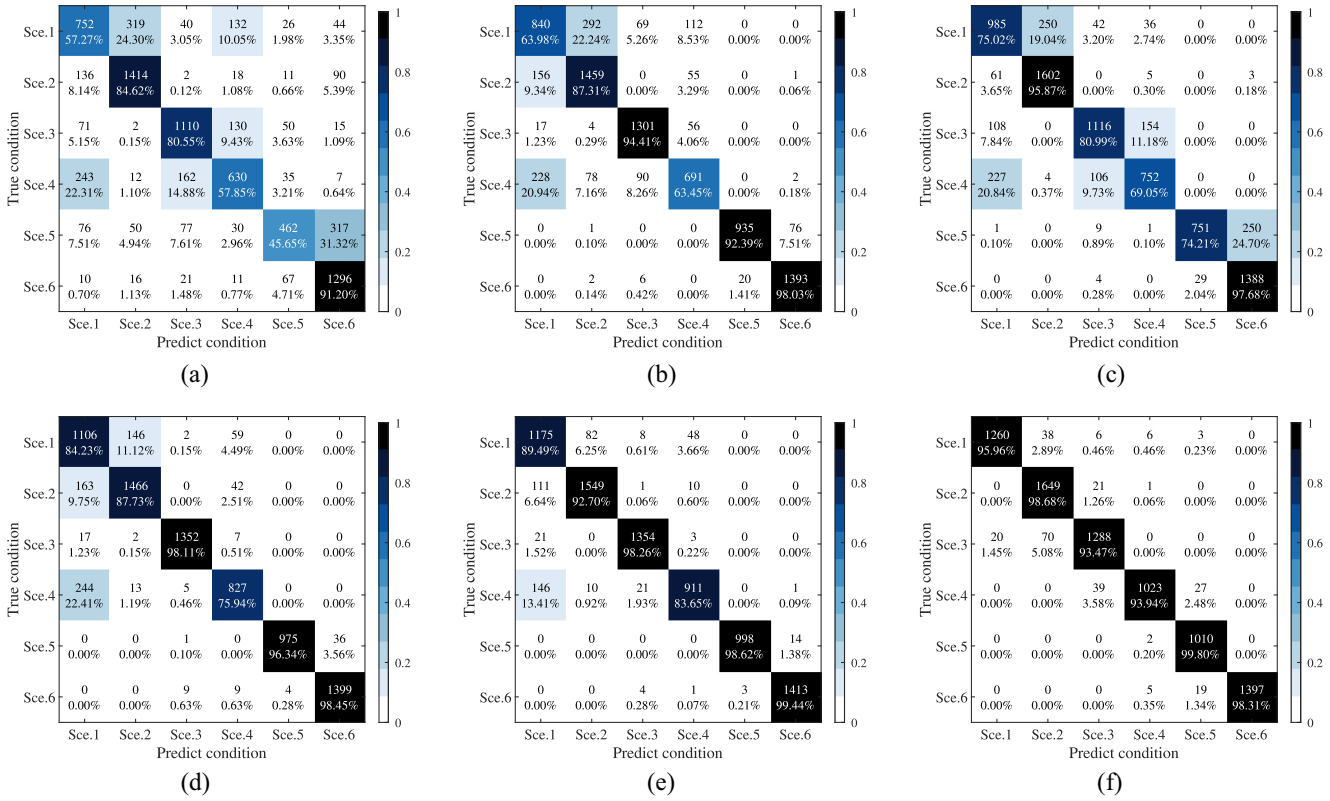


Fig. 9. Confusion matrix results for multiclassification of scenarios Sec.1–6: (a) LSTM, (b) CNN+LSTM, (c) CNN-A, (d) MIL, (e) PSTFFN-W, and (f) PSTFFN.

TABLE VII
ERROR CORRECTION EFFECT IN DIFFERENT SCENARIOS

Scenario	MAE(m)		RMSE(m)		time(ms)
	Original	Mitigated	Original	Mitigated	
Sec.1	0.0598	0.0267	0.0647	0.0318	0.445
Sec.2	0.1907	0.0335	0.1660	0.0391	0.432
Sec.3	0.1565	0.0309	0.1874	0.0468	0.456
Sec.4	0.5791	0.0783	0.5499	0.0884	0.439
Sec.5	1.8397	0.1174	2.3911	0.0956	0.423
Sec.6	3.5858	0.1799	6.7233	0.1508	0.451
Overall	1.0686	0.0778	1.6804	0.0754	0.441

occlusion scenario of Fig. 11(b), the CDF for the direct method (blue curve) even exceeds the original error's CDF in 95% of cases ($Y = 95\%$, where the x-value of the blue curve exceeds that of the red curve), indicating increased error. This is due to the significant differences in distance error distributions across various NLOS occlusion types, making it challenging for the direct STARN method to capture effective features. Therefore, this study employs PSTFFN for NLOS identification before applying targeted error prediction models for specific scenarios, substantially enhancing error suppression performance.

4) Indoor Localization Performance: For the purpose of evaluating the effectiveness of our NLOS detection and mitigation approach in real UWB indoor localization scenarios, we conducted localization experiments in the hallway and underground parking garage environments shown in Fig. 7(a) and (b). In each scenario, the four anchor points were positioned at fixed locations, measured in advance using the Leica Disto X3, while the tag continuously moved along a reference path. The previously trained PSTFFN and STARN model from the NLOS detection and error mitigation experiments were used for testing localization performance. To ensure fair testing, the location of the tag and the anchor arrangements differed as much as possible from those used in the training data shown in Fig. 7(a) and (b).

We compared our proposed method with LS, EKF, and an approach that applies STARN directly for error mitigation without NLOS subdivision. To demonstrate performance more clearly, the LS+EKF approach incorporated an MPU9250 IMU to obtain the tag's kinematic state and corrected predictions using UWB measurements. For other comparison methods, only UWB distance measurements from the DWM1000 were used. The localization results comparison for various methods across the two scenarios in Fig. 12 highlights that using the proposed PSTFFN and STARN combination for range error mitigation, followed by LS for localization, yields the smallest error (with the green curve closely following the reference trajectory in black). Compared to the unprocessed method that directly applies LS for localization (red curve), the proposed method significantly reduces the accuracy fluctuations caused by transitions between NLOS environments.

As obtaining the real-time position of a moving tag and calculating its errors is challenging, markers were placed every 0.1 m along the reference trajectory, and the tag was positioned at each marker to measure the error for each positioning method relative to these reference positions. Fig. 13

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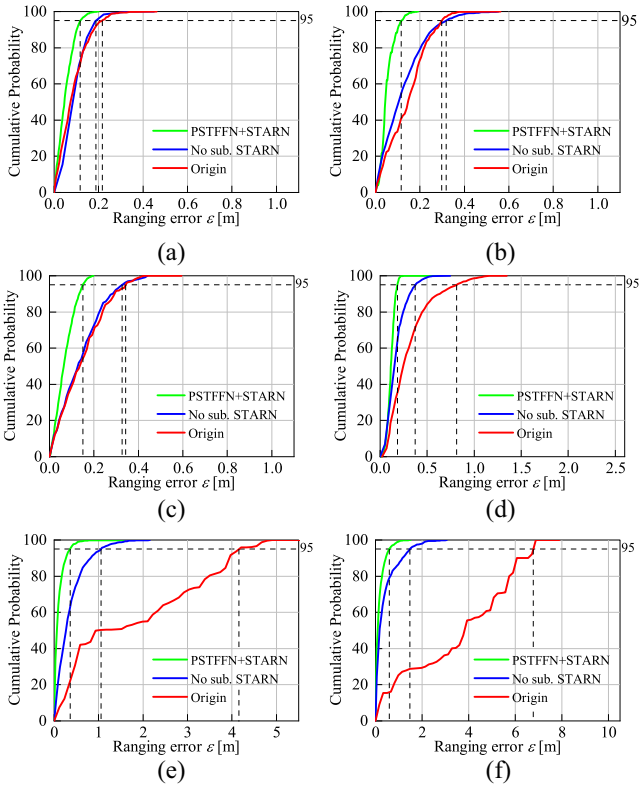


Fig. 11. Comparison of the CDFs between Post-NLOS classification mitigation and direct error mitigation methods, (a) Sec.1, (b) Sec.2, (c) Sec.3, (d) Sec.4, (e) Sec.5, and (f) Sec.6.

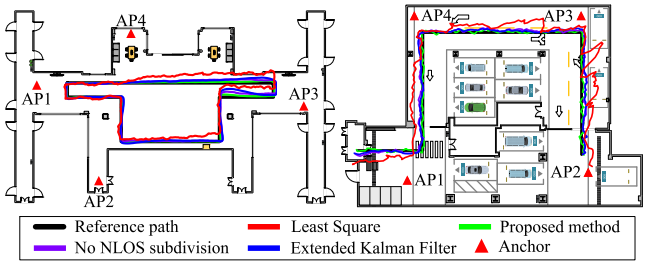


Fig. 12. Comparison of dynamic localization experiments in two indoor scenarios.

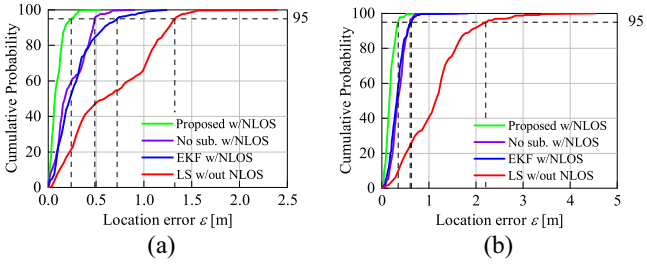


Fig. 13. CDF of localization in different scenarios. (a) Hallway. (b) Underground Parking Garage.

presents the CDF of positioning errors for each approach in the hallway and underground parking garage scenarios. Both comparison methods mitigate positioning errors to a certain extent, as evidenced by their 95% CDF values being below those of the LS method (red curve). The proposed method, which first applies error mitigation before using LS for positioning, demonstrates a clear performance advantage, achieving a positioning error below 0.24 m for 95% of cases in the hallway scenario.

In comparison, the approach using STARN for error mitigation before LS positioning (purple curve) maintains positioning errors below 0.49 m in 95% of Hallway cases, while LS+EKF achieves an error of under 0.72 m (blue curve). The performance of LS+EKF and STARN is relatively close, though LS+EKF exhibits a slight advantage in the 64% to 74% interval on the y-axis, as indicated by lower x-values in Fig. 13(a) within this range. A similar trend is observed in the Underground Parking Garage, where the STARN-only error mitigation and LS+EKF methods perform almost identically, whereas the proposed method achieves higher accuracy for 95% of cases. This result could be due to limitations in IMU accuracy affecting LS+EKF performance. However, it also demonstrates the cost-effectiveness of our method, as it achieves higher accuracy than standard UWB/IMU fusion systems using only UWB sensors.

TABLE VIII
COMPARISON OF LOCALIZATION PERFORMANCE OF VARIOUS METHODS IN DIFFERENT SCENARIOS

Scenario	Methods	MAE(m)	RMSE(m)	time(ms)
Hallway	LS	0.675	0.805	0.104
	LS+EKF	0.291	0.363	0.163
	No NLOS subdivision	0.241	0.293	0.541
	PSTFFN+ STARN	0.094	0.117	1.630
Parking	LS	1.157	1.314	0.110
	LS+EKF	0.333	0.369	0.165
	No NLOS subdivision	0.358	0.387	0.549
	PSTFFN+ STARN	0.180	0.206	1.648

Additionally, Table VIII presents the MAE and RMSE of positioning errors, along with the computational time per localization for each method after receiving UWB signals, with all results averaged over 100 test trials. From these results, it is evident that the proposed PSTFFN+STARN combination approach, although computationally more demanding with processing times of 1.630 ms and 1.648 ms, achieves the lowest MAE (0.094 m in the hallway scenario and 0.180 m in the underground parking garage) and RMSE (0.117 m in the hallway scenario and 0.206 m in the underground parking garage). Compared to the method that directly uses STARN for error mitigation (with MAE values of 0.241 m and 0.358 m and RMSE values of 0.293 m and 0.387 m in the hallway and underground parking garage, respectively), the proposed PSTFFN+STARN combination method demonstrates a clear advantage in reducing positioning errors. The average time per localization is less than 2 ms (excluding transmission and data transfer delays), which is sufficient to meet the requirements of most personnel and vehicle localization tasks. Furthermore, compared to the traditional LS+EKF approach, the PSTFFN+STARN combination method shows superior performance, especially in challenging NLOS environments.

V. CONCLUSION

In this article, we introduce a novel NLOS detection and distance error mitigation approach using a PSTFFN. By

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transforming CIR data into 2-D TFDOCIR image data and integrating manually extracted features as auxiliary inputs, this approach fully leverages the potential of UWB CIR for NLOS recognition and error mitigation without requiring any prior knowledge. Using the NLOS detection results, the approach can select an appropriate STARN model to predict and correct NLOS distance measurement errors. Experimental results indicate that PSTFFN outperforms traditional ML approaches in the standard LOS/NLOS binary classification task, achieving an accuracy of 97.10%. To further validate its performance, we tested the proposed method on a self-built NLOS dataset with common indoor obstacles, and PSTFFN demonstrated excellent multiclass classification capabilities with an accuracy of 96.74%. For NLOS distance error mitigation, we directly compared STARN-based predictions across various NLOS scenarios. Results show that first using PSTFFN for fine categorization, followed by the targeted STARN model for error mitigation, yields superior results. This combined approach lowers the RMSE and MAE of distance errors from 1.6804 m and 1.0686 m to 0.0754 m and 0.0778 m, respectively. In indoor positioning experiments, this approach improved localization accuracy by more than 81.86%, completing error mitigation and localization within 1.648 ms after receiving the UWB signals. Future research, we will further explore the application of this model across varied environments, enhance the accuracy of the distance error prediction model, and attempt to deploy it on embedded devices.

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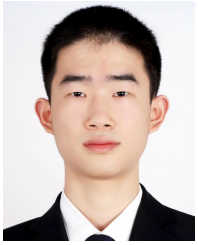
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